5) $F(X,Y,Z) = (X+\overline{Y})\cdot(Y+Z)\cdot\overline{X\cdot Z}$

X	Y	Z	$\bar{y}$	$x+\bar{y}$	$(y+z)$	$(\overline{x\cdot z})$	F
0	0	0	1	1	0	1	0
0	0	1	1	1	1	1	1
0	1	0	0	0	1	1	0
0	1	1	0	0	1	1	0
1	0	0	1	1	0	1	0
1	0	1	1	1	1	0	0
1	1	0	0	1	1	1	1
1	1	1	1	1	1	0	0

$\bar{y}\ \bar{y}z + xy\bar{z} + xyz$

$\bar{y}\ \bar{y}z + xy\bar{z}$

$= \overline{\bar{x}\bar{y}z + xy\bar{z}}$

$= (\overline{\bar{x}\bar{y}z})\cdot(\overline{xy\bar{z}})$

$= (x+y+\bar{z})\cdot(\bar{x}+\bar{y}+z)$

$= x\bar{x} + \underline{x\bar{y}} + \underline{xz} + \underline{\bar{x}y} + \underline{y\bar{y}} + \underline{yz} + \bar{z}\bar{x} + \underline{\bar{z}y} + \cancel{\bar{z}z}$

$= 0\ \ x\bar{y} + y\bar{y} + \bar{x}y + \bar{z}y + yz + yz + \bar{x}\bar{z}$